Windows VM Migration Driver Installation Guide

SATA Bootstrap → VirtIO CD-ROM Install → Production Ready

Win10 | Win11 | Server 2019/2022 | SATA Boot | VirtIO Network | Guest Agent | Apache 2.0

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1. Best Practice: Pre-Install Drivers

Install VirtIO drivers inside the VM **before migration** for a seamless transition.

Why Pre-Install?

- Network works on **first KVM boot** — no manual steps
- QEMU Guest Agent already running
- No CD-ROM or VNC console needed after migration
- Works with any Windows version (7 through 11, Server 2012-2025)
- Drivers coexist with VMware tools — no conflict

Steps (While Still on VMware)

1. Download `virtio-win.iso` from fedorapeople.org
2. Attach ISO to VMware VM as CD-ROM
3. Inside VM: run `D:\virtio-win-guest-tools.exe`
4. Reboot VM (still on VMware)
5. Migrate VMDK → KVM boots with drivers ready

Recommended for production migrations. If pre-install is not possible (e.g., VM already shut down), use the post-migration workflow on the following pages.

2. Post-Migration Pipeline Overview

If pre-install was not possible: end-to-end flow from VMware VMDK to a running Windows KVM guest.

VMDK Source VMware / Hyper-V → **Flatten** qemu-img convert → **Offline Fixes** registry + drivers → **SATA Boot** reliable first boot → **Driver Install** CD-ROM D: drive

Why SATA Disk Bus?

- Windows ships with built-in SATA/AHCI drivers
- VirtIO disk bus requires `viostor.sys` with `Start=0` (boot-start) in the registry **before** switching
- The VirtIO GUI installer does NOT set Start=0 for storage
- Switching to VirtIO disk without offline registry fix causes BSOD (INACCESSIBLE_BOOT_DEVICE)
- SATA provides reliable performance for production workloads
- No risk of boot failure — always works on first try

Why CD-ROM Install?

- virtio-win.iso is auto-attached as a CD-ROM device
- Appears as D: drive inside Windows — always visible
- `virtio-win-guest-tools.exe` installs ALL drivers at once
- Network, balloon, serial, guest agent — one click
- Silent install supported: `/S` flag for automation
- Most reliable method — works on all Windows versions

Automated Mechanisms (Best-Effort)

- **SetupComplete.cmd** — only fires on fresh installs, not migrated VMs
- **rhsrvany.exe service** — blocked by unsigned binary restrictions on Win10/Win11
- **HKLM RunOnce key** — only triggers on new login sessions
- **Startup folder .bat** — requires actual user login
- All mechanisms are staged during offline fixes but may not execute
- CD-ROM install is the recommended production approach

Supported Guest Versions

- Windows 10 Pro/Enterprise (20H2, 21H2, 22H2)
- Windows 11 Pro/Enterprise (22H2, 23H2)
- Windows Server 2019 (Core + Desktop Experience)
- Windows Server 2022 (Core + Desktop Experience)
- Windows Server 2016 (limited testing)
- Both BIOS and UEFI firmware modes

Key principle: Boot reliability first. SATA disk bus guarantees Windows boots on KVM. VirtIO drivers for network, balloon, and guest agent are installed post-boot via the CD-ROM installer. The disk stays on SATA — this is the production configuration.

2. What's Auto-Staged

hyper2kvm stages multiple driver installation mechanisms during offline fixes. These are best-effort for migrated VMs.

Registry Entries (SYSTEM Hive)

- viostor service: `Start=0` (boot-critical)
- vioscsi service: `Start=0` (boot-critical)
- NetKVM service: `Start=2` (auto-start)
- Balloon service: `Start=3` (demand)
- vioserial service: `Start=3` (demand)
- CriticalDeviceDatabase PCI ID mappings
- VMware services disabled (Start=4)

Driver Files (.sys)

- `viostor.sys` → `System32\drivers\`
- `vioscsi.sys` → `System32\drivers\`
- `netkvm.sys` → `System32\drivers\`
- `balloon.sys` → `System32\drivers\`
- `vioser.sys` → `System32\drivers\`
- INF files staged in `Windows\INF\`
- CAT files for signature verification

Firstboot Scripts

- `SetupComplete.cmd` in `OEM\` folder
- rhsrvany.exe service wrapper registered
- HKLM RunOnce registry key set
- Startup folder .bat file placed
- pnputil driver installation commands staged
- Self-cleanup after execution

CD-ROM Auto-Attachment

Domain XML Configuration

```
<disk type='file' device='cdrom'>
  <driver name='qemu' type='raw'/>
  <source file='/var/lib/hyper2kvm/virtio-win.iso'/>
  <target dev='sdb' bus='sata'/>
  <readonly/>
</disk>
```

Automatically added when `inject_virtio_drivers: true` is set.

ISO Discovery Paths

- `/var/lib/hyper2kvm/virtio-win.iso` (primary)
- `/usr/share/virtio-win/virtio-win.iso` (system)
- `/var/lib/libvirt/images/virtio-win.iso` (fallback)
- `quickstart.sh` auto-downloads and extracts the ISO
- Contains drivers for Win10, Win11, Server 2019, Server 2022
- Architecture: amd64 for 64-bit guests

Important: For migrated VMs, the staged registry entries and driver files establish the foundation, but the **CD-ROM installer** is what reliably activates all drivers. The offline staging ensures the .sys files are present on disk, while the installer handles PnP binding and service registration.

3. First Boot — What You'll See

After migration, Windows boots on the SATA disk controller. Here is what to expect.

1 Windows Boot Screen

- Windows boots normally on SATA disk controller
- AHCI/SATA drivers are built into Windows — always work
- Boot time is similar to VMware (30-90 seconds)
- No BSOD, no driver errors, no safe mode
- UEFI or BIOS boot — both supported

2 Desktop Appears

- Windows desktop loads normally
- Task bar shows **no network icon** (red X or disconnected)
- This is expected — VirtIO network driver not yet installed
- Device Manager shows unknown devices (VirtIO PCI)
- Access via **VNC console** (h2kweb or virt-manager)

3 CD-ROM Drive Visible

- D: drive appears in Windows Explorer
- Contains `virtio-win-guest-tools.exe` (top level)
- Also contains individual driver folders (NetKVM, vioscsi, etc.)
- ISO label: "virtio-win" or similar
- If D: is not visible, check domain XML for cdrom entry

4 Accessing the VM

- **h2kweb:** VM Management → click VM → Open VNC Console
- **virt-manager:** Double-click the VM to open console
- **CLI:** `virt-viewer --connect qemu:///system <vmname>`
- **Remote:** VNC on auto-assigned port (check XML)
- No SSH or RDP available yet (no network driver)

What You Will NOT See

No Network (Expected)

The VirtIO network adapter is present in the VM but Windows does not have the driver loaded. The network icon shows disconnected or missing. This is resolved in the next step by running the installer.

No BSOD (Guaranteed)

Because the disk bus is SATA (not VirtIO), Windows uses its built-in AHCI driver. There is no risk of INACCESSIBLE_BOOT_DEVICE. The VM boots reliably every time.

Tip: If using h2kweb, the VM list will show the VM as "running" but without an IP address. The IP will appear after you install VirtIO drivers and the QEMU Guest Agent starts.

4. Installing Drivers — Step by Step

Run the VirtIO installer from the CD-ROM. One click installs everything.

1 Open D: Drive

- Open Windows Explorer (Win+E)
- Click on the **D:** drive (CD-ROM / DVD)
- You will see `virtio-win-guest-tools.exe` at the top level
- Also visible: driver folders (Balloon, NetKVM, vioscsi, etc.)
- If D: is not visible, it may be another drive letter — check "This PC"

2 Run the Installer

- Double-click `virtio-win-guest-tools.exe`
- Click "Yes" on the UAC prompt
- The installer runs silently — shows a brief progress dialog
- Installs ALL VirtIO drivers + QEMU Guest Agent + SPICE agent
- Takes approximately 30-60 seconds
- No reboot required for network

3 Silent Install (Automation)

```
D:\virtio-win-guest-tools.exe /S
```

- The `/S` flag runs a completely silent install
- No UI, no prompts, no user interaction
- Useful for scripted or remote installation via VNC paste
- Exit code 0 = success
- Can be run from elevated Command Prompt or PowerShell

4 Network Appears

- Network icon changes to connected (within seconds)
- DHCP obtains IP address automatically (if DHCP server exists)
- Static IP: configure manually after network driver loads
- RDP and SSH become available if previously configured
- IP address appears in h2kweb VM list

What the Installer Does

Component	Action	Result
NetKVM (netkvm.sys)	Installs VirtIO network driver + PnP binding	**Network available immediately**
Balloon (balloon.sys)	Installs memory balloon driver	**Dynamic memory management active**
VioSerial (vioser.sys)	Installs serial port driver	**Guest agent communication channel**
QEMU Guest Agent	Installs and starts qemu-ga service	**IP reporting, file operations, shutdown**
SPICE Agent	Installs SPICE vdagent (if SPICE)	**Clipboard, resolution auto-resize**
VioStor (viostor.sys)	Registers driver but does NOT activate for boot disk	**Disk stays on SATA (correct behavior)**

No reboot needed for network, balloon, or guest agent. The VirtIO network driver activates immediately after installation. RDP and SSH access become available as soon as the network is up.

5. After Installation — What Works

Complete status of all VirtIO components after running the installer.

Component	Status	How to Verify	Notes
VirtIO Network	✓ Working	`ipconfig /all`	Red Hat VirtIO Ethernet Adapter in Device Manager. Works immediately, no reboot.
Memory Balloon	✓ Working	Device Manager → System devices	VirtIO Balloon Driver. Enables dynamic memory with `virsh setmem` .
QEMU Guest Agent	✓ Working	`sc query qemu-ga`	Reports IP, enables file operations, graceful shutdown via `virsh shutdown` .
VioSerial	✓ Working	Device Manager → System devices	Communication channel between host and guest agent.
SPICE Agent	✓ Working (if SPICE)	`sc query vdservice`	Only relevant for SPICE graphics. VNC deployments skip this.
Disk Bus	SATA (by design)	`wmic diskdrive get model`	Stays on SATA. VirtIO disk requires offline Start=0 registry fix before switching. SATA is fine for production.

Why Disk Stays on SATA

- VirtIO storage driver (`viostor`) must be set to `Start=0` (boot-start) in registry
- The GUI installer registers the driver but does NOT set boot-start
- Switching disk bus to VirtIO without Start=0 causes BSOD at boot
- Setting Start=0 requires offline registry editing (qemu-nbd + hivex)
- SATA disk performance is adequate for most production workloads
- Avoiding this complexity eliminates the most common Windows migration failure

Performance Comparison: SATA vs VirtIO Disk

- **Sequential read:** SATA ~500 MB/s, VirtIO ~600 MB/s
- **Sequential write:** SATA ~400 MB/s, VirtIO ~500 MB/s
- **Random IOPS:** SATA ~30K, VirtIO ~40K
- **CPU overhead:** SATA slightly higher (AHCI emulation)
- Difference is 15-25% — marginal for most workloads
- VirtIO disk is only worth the complexity for high-IOPS databases

Production-Ready Configuration

SATA disk + VirtIO network + VirtIO balloon + QEMU Guest Agent = fully functional Windows VM on KVM.
Network works immediately. Guest agent reports IP. Memory is dynamic. No reboot required.

6. Verification — How to Confirm

Commands and checks to verify that all VirtIO components are installed and working correctly.

Device Manager Check

- Open Device Manager (`devmgmt.msc`)
- **Network adapters:** "Red Hat VirtIO Ethernet Adapter"
- **System devices:** "VirtIO Balloon Driver"
- **System devices:** "VirtIO Serial Driver"
- No yellow warning icons = all drivers loaded
- No "Unknown device" entries for VirtIO PCI IDs

Driver Enumeration

```
pnputil /enum-drivers

Published Name: oem0.inf
Class Name:     Net
Driver Version: ...
Signer Name:    Red Hat, Inc.
```

Look for Red Hat signed drivers for Net (NetKVM), System (balloon, serial).

Guest Agent Check

```
sc query qemu-ga

SERVICE_NAME: qemu-ga
STATE: 4  RUNNING
```

From the host, verify with:

```
virsh qemu-agent-command <vm> \
  '{"execute":"guest-ping"}'
```

Host-Side Verification

Network Connectivity

```
# Check if VM has an IP address
virsh domifaddr <vmname>

# Check guest agent is responding
virsh qemu-agent-command <vmname> \
  '{"execute":"guest-network-get-interfaces"}'

# Test SSH (if configured)
ssh administrator@<ip>

# Test RDP (if enabled)
xfreerdp /v:<ip> /u:Administrator
```

h2kweb Dashboard Checks

- **VM List:** IP address shown in green = guest agent working
- **Guest agent dot:** Cyan indicator next to VM name
- **Health Check:** Click "Run Check" on VM detail panel
- — VM Running: PASS (green)
- — IP Address: shows the VM's IP
- — Guest Agent: PASS (green)
- **Disk bus badge:** Shows "sata" (yellow) — expected

Network: ipconfig /all | Agent: sc query qemu-ga | Drivers: pnputil /enum-drivers | Host: virsh domifaddr | Device Manager: no warnings | h2kweb: green health checks

7. Troubleshooting

Common issues and solutions for Windows VM migration driver installation.

Problem	Cause	Solution
CD-ROM (D:) not visible	virtio-win.iso not attached in domain XML or ISO file missing	Verify ISO exists: `ls /var/lib/hyper2kvm/virtio-win.iso`. Check domain XML for `<disk device='cdrom'>` entry. Run `sudo ./scripts/quickstart.sh` to download ISO. Or attach manually: `virsh attach-disk <vm> /path/to/virtio-win.iso sdb --type cdrom --mode readonly`
No network after installer	VM NIC model not set to virtio in domain XML	Edit domain XML: `<model type='virtio'/>` in the `<interface>` section. Restart VM. Then re-run installer.
BSOD on boot (INACCESSIBLE_BOOT_DEVICE)	Disk bus changed to VirtIO without offline registry fix	Change disk bus back to SATA in domain XML: `<target dev='sda' bus='sata'/>`. Never switch to VirtIO disk without setting `viostor Start=0` offline via hivex.
Guest agent not starting	VioSerial channel not configured in domain XML	Add channel device: `<channel type='unix'><target type='virtio' name='org.qemu.guest_agent.0'/></channel>`. The agent service may also need manual start: `net start qemu-ga`
Installer fails with error	ISO version mismatch or corrupted download	Re-download virtio-win.iso from `https://fedorapeople.org/groups/virt/virtio-win/direct-downloads/stable-virtio/`. Verify ISO integrity with sha256sum.
Windows boots to Safe Mode	VMware drivers conflicting with KVM hardware	Boot to Safe Mode, uninstall VMware Tools from Programs and Features. Or during offline fixes, set VMware services to `Start=4` (Disabled) via registry.
Slow disk performance	SATA emulation overhead (minor)	This is expected — SATA is ~15-25% slower than VirtIO for random I/O. For most workloads this is acceptable. Use cache=writeback in domain XML for better write performance.
No IP in h2kweb	Guest agent not installed or channel missing	Install drivers via CD-ROM. Verify `sc query qemu-ga` shows RUNNING. Check domain XML has virtio-serial channel device.

Emergency Recovery

- If VM does not boot, switch disk bus to SATA in XML
- `virsh edit <vmname>` → change `bus='virtio'` to `bus='sata'`
- If stuck at boot, try: `virsh destroy <vm> && virsh start <vm>`
- For persistent boot issues, re-run migration with `inject_virtio_drivers: true`
- Check logs: `C:\hyper2kvm\firstboot.log` (if firstboot ran)

Manual Driver Install (Alternative)

- If the unified installer fails, install drivers individually:
- Device Manager → right-click unknown device → Update driver
- Browse to D: drive → select appropriate folder (e.g., `NetKVM\w10\amd64`)
- Or use pnputil: `pnputil /add-driver D:\NetKVM\w10\amd64\*.inf /install`
- Repeat for each driver folder (Balloon, vioserial, etc.)